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TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

PROJECT PROPOSAL

SUBJECT: INTRODUCTION TO SOFTWARE ENGINEERING

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2 Introduction and Motivation

Managing rental properties involves many connected activities: keeping room and tenant records, maintaining rental contracts, recording electricity and water usage, calculating monthly charges, tracking payments and deposits, handling maintenance requests, and communicating with tenants. When these activities are recorded manually or spread across separate files and messaging channels, it can be difficult to maintain consistent information and quickly determine the current status of a property.

The Trozy is proposed as a web application that brings these activities together in one place. It is designed for two business roles only: Landlord and Tenant. A landlord can manage multiple rental properties and their rooms, while tenants can use a dedicated portal to view information related to their own room, contract, invoices, payments, and requests.

The system also has a roadmap for intelligent features, including utility-usage analysis, AI-assisted questions, optical character recognition (OCR), maintenance-request classification, and forecasting. These features are intended to support users rather than replace their decisions. The core rental-management workflows should remain reliable before advanced AI features are introduced.

2.1 Problem Statement

Rental management requires accurate coordination of information throughout the rental lifecycle. The main problems addressed by this project are:

- **Fragmented records:** Property, room, tenant, contract, invoice, and maintenance information may be kept in different places, making it harder to retrieve and update.
- **Manual billing work:** Landlords need to record meter readings, apply utility rates and service fees, calculate charges, issue invoices, and track due dates.
- **Limited payment visibility:** Tenants need to know what they owe and what they have paid, while landlords need to distinguish received payments, outstanding debts, partial payments, and overdue invoices.
- **Difficult handover and deposit settlement:** Room condition, included assets, final utility readings, outstanding charges, and deposit deductions must be recorded when a tenant moves in or out.
- **Slow maintenance communication:** Repair requests need descriptions, evidence, status updates, and a record of the final resolution.
- **Limited operational overview:** Landlords need a consolidated view of occupancy, revenue, expenses, debts, contracts approaching expiry, and unresolved incidents.

- **Data access and privacy:** Each landlord must only manage properties under their control, and each tenant must only access information they are authorized to see.

The project addresses these problems by consolidating the relevant workflows and applying role-based access rules across the system. The supplied project material does not include measured data about current landlords' workflows, so the scale and frequency of these problems should be validated through interviews.

2.2 Proposed Solution

The proposed solution is a web-based application organized around the rental-management lifecycle. Its main capabilities are:

1. **Property and tenancy management:** Manage properties, rooms, tenant profiles, room members, rental contracts, contract renewals, and contract termination.
2. **Billing and finance:** Record electricity and water readings, configure services and additional fees, generate invoices, record payments, track debts, manage deposits, and record income and expenses.
3. **Tenant self-service:** Provide a tenant portal for room and contract information, invoices, payment history, utility usage, notifications, and requests.
4. **Operations and communication:** Handle maintenance incidents, assets and equipment, move-in/move-out handovers, announcements, requests, documents, house rules, vehicles, and shared amenities.
5. **Dashboards and configuration:** Summarize occupancy, revenue, outstanding debts, expenses, expiring contracts, and unresolved incidents; allow landlords to configure billing rules and manage data.
6. **Intelligent assistance roadmap:** Explore utility anomaly detection, OCR-assisted meter-reading and document entry, AI-supported maintenance classification, natural-language operational queries, and longer-term forecasting.

The system has two roles. Landlords manage their own properties and the associated operational data. Tenants view and interact only with the records related to their own tenancy and authorized information. In the initial version, bank transfers may be recorded and confirmed by the landlord; pressing a "transferred" button must not automatically be treated as proof that payment has been received.

2.3 Why the Project Is Worth Developing

The project is worth developing because it targets a set of recurring activities that are closely connected but often require repeated manual work. Centralizing them can make rental information easier to maintain and retrieve, improve visibility into billing and

payment status, and create a consistent record of maintenance and handover events.

The system can also improve transparency between landlords and tenants. Tenants can inspect invoice details, utility readings, payment history, and request status rather than relying entirely on separate conversations. Landlords can use consolidated reports to monitor room occupancy, cash received, outstanding debts, expenses, and upcoming contract deadlines.

A modular design makes the project suitable for an introductory software-engineering course. Core modules can be developed and tested first, while AI, OCR, and forecasting remain optional extensions. This approach allows the team to demonstrate a complete, useful workflow without assuming that advanced features will be accurate before sufficient data and validation are available.

The project should not be presented as a rental marketplace. Its agreed scope excludes public room listings, room searching, viewing appointments, brokers, and employee roles. Its purpose is to manage an existing landlord–tenant relationship.

3 Target Users and Operating Environments

3.1 Target Users

The system has two target user groups.

Landlord

A landlord may manage one or more rental properties and multiple rooms under the same account. Typical tasks include:

- Creating and updating property and room records.
- Maintaining tenant profiles and linking tenants to the correct room and contract.
- Creating, renewing, and ending rental contracts.
- Recording electricity and water readings and configuring service charges.
- Generating invoices and confirming payments.
- Tracking debts, deposits, income, and expenses.
- Handling repair requests and recording equipment history.
- Managing move-in and move-out handovers.
- Sending announcements and reviewing reports.

Tenant

A tenant uses a dedicated portal to access information relevant to their tenancy. Typical tasks include:

- Viewing room, contract, house-rule, and deposit information.
- Checking invoices, due dates, utility readings, and payment history.
- Reviewing notifications from the landlord.
- Submitting repair requests with descriptions and photos.
- Following the progress of requests and confirming their resolution.
- Requesting contract renewal or move-out.
- Accessing documents and handover records they are permitted to view.

A landlord may create a tenant profile before the tenant registers. The tenant can then be invited or linked through a code so that the account connects to the correct tenant profile, room, and contract.

3.2 Environments and Accessibility

The system is planned as a web application, so a separate native mobile application is not required by the agreed scope. Users should be able to access the system through a web browser on a computer, tablet, or smartphone, provided that the implemented interface supports the device and browser.

The interface should present different functions according to the signed-in user's role. Landlords need management screens for properties, rooms, contracts, billing, operations, and reports. Tenants need a simpler self-service portal focused on their own tenancy, charges, payments, documents, and requests.

Accessibility and usability considerations include:

- Use clear labels, consistent navigation, readable text, and understandable status indicators.
- Keep billing calculations and invoice line items visible so that users can understand how totals are formed.
- Provide clear feedback when a form is saved, invalid, or incomplete.
- Support photo and document uploads where required by meter-reading, maintenance, and handover workflows.
- Make key pages usable on smaller screens, particularly invoice viewing and maintenance-request submission.
- Protect personal and financial records through authentication, authorization, and appropriately limited access.

- Keep a history of important changes to readings, invoices, contracts, and other operational records.

These are design goals for implementation, not claims that accessibility testing or security verification has already been completed. Specific browser support, accessibility standards, hosting arrangements, backup frequency, and data-retention rules remain to be confirmed during requirements specification.

4 Key Features of the Application

The system's functionality is grouped into five business areas.

4.1 Account, Property, Room, Tenant, and Contract Management

- Register, sign in and out, recover passwords, verify email, and manage profile details.
- Maintain room codes, area, rental price, capacity, amenities, photos, and room status.
- Maintain tenant profiles, room members, move-in/move-out dates, and rental history.
- Create contracts from templates, record rent and deposit amounts, manage dates and terms, renew or terminate contracts, and retain contract versions and appendices.

4.2 Utilities, Billing, Payments, Deposits, and Expenses

- Record electricity and water meter readings by billing period, including meter photos and reading history.
- Configure utility rates and service fees such as internet, cleaning, waste collection, parking, and management fees.
- Generate periodic invoices containing rent, utilities, services, and other adjustments.
- Track invoice due dates and statuses, including unpaid, partially paid, and fully paid.
- Record cash and bank-transfer payments, provide transfer details or VietQR information, issue receipts, and track overdue debts.
- Record deposits, deposit additions, deductions, refunds, and settlement evidence.
- Record income and operating expenses, classify expenses, and produce monthly

summaries and exports.

4.3 Tenant Portal and Property Operations

- Give tenants a dashboard showing their current room, contract, deposit, latest invoices, outstanding amounts, payment history, and utility usage.
- Let tenants submit maintenance requests with a category, description, urgency, and supporting photos.
- Let landlords update repair status, expected repair time, costs, and evidence of completion.
- Track room assets and equipment, including condition, photos, purchase or warranty details when available, and repair history.
- Send announcements and reminders for bills, contract expiry, maintenance, and utility interruptions.

4.4 Dashboards, Reports, and Data Configuration

- Summarize total rooms, occupied and vacant rooms, and occupancy rate.
- Report revenue by month or property, outstanding debts, overdue invoices, income, expenses, and estimated operating results.
- Show contracts approaching expiry and unresolved maintenance incidents.
- Allow landlords to configure billing dates, invoice periods, service rates, notification rules.
- Maintain records of important operational changes.

Tenants should see only their own relevant costs, payments, and utility history. Landlord-level financial and property reports must not be exposed to tenants.

4.5 Smart Utility Insights and AI Assistance

The planned intelligent capabilities include:

- Comparing utility consumption across billing periods and flagging unusual increases or decreases.
- Detecting meter readings that violate configured validation rules.
- Using OCR to suggest readings from meter photos and extract selected fields from documents.
- Exploring forecasts for debts, revenue, and vacant rooms when enough reliable historical data is available.

These capabilities are proposed extensions. OCR readings must be checked and confirmed by the landlord before being used for billing. AI outputs should support, not independently make, decisions such as deposit deductions, contract changes, or payment confirmation. An unusual utility reading is a reason to investigate, not proof of a leak or equipment failure.

5 Typical User Journey

The following journey describes a typical rental lifecycle using the modules in the supplied project material.

Step 1: Set up a property and room

The landlord creates a property record, adds its rooms, enters room details and rental prices, and configures relevant services and house rules. The landlord can then see which rooms are vacant or occupied.

Step 2: Register a tenant and create a contract

The landlord creates a tenant profile and assigns the tenant to a room. A rental contract is prepared with the rental price, deposit, start and end dates, payment schedule, and terms. The landlord records the deposit and prepares a move-in handover record, including the room's condition, assets, and initial utility readings.

Step 3: Give the tenant portal access

The tenant receives an invitation or linking code and connects their account to the correct tenant profile and contract. After signing in, the tenant can view permitted room details, contract information, deposit records, house rules, and relevant documents.

Step 4: Record utilities and issue a monthly invoice

At the end of a billing period, the landlord records electricity and water meter readings and checks the evidence. The system calculates consumption using the configured rates, adds applicable services and other charges, and prepares an invoice for review. After the landlord checks and issues the invoice, the tenant receives a notification and can inspect the line items and due date.

Step 5: Pay the invoice and update the balance

The tenant reviews the amount due and uses the provided payment information, such as bank-transfer details or a VietQR code. The landlord confirms receipt in the basic

version of the system. The payment history and remaining debt are updated accordingly, including cases of partial payment or overdue balances.

Step 6: Submit and resolve a maintenance request

If an issue occurs, the tenant submits a request with a description and, where useful, a photo. The landlord reviews it, updates the status and expected repair time, and records the work and costs. The tenant receives status updates and can confirm whether the issue has been resolved. AI-assisted classification may be added later to help categorize the request.

Step 7: Request move-out and settle the tenancy

When the tenant requests to leave, the landlord and tenant inspect the room and its assets and record the final condition. The landlord closes the utility period, calculates outstanding charges, reviews any damage evidence, and determines the deposit deduction or refund. The tenancy record is closed and the room status is updated when the handover and settlement process is complete.

Step 8: Review operational performance

The landlord reviews dashboards and reports to understand occupancy, collected revenue, outstanding debts, expenses, expiring contracts, and unresolved maintenance incidents. Utility insights may flag unusual consumption, while AI-assisted queries may help retrieve relevant records. Any flagged issue or forecast should be checked against the underlying records before action is taken.

6 Interviews

Interview status: The source material supplied for this report describes the proposed system and its functions, but it does not contain interview transcripts, participant details, or verified interview findings. Therefore, the following is an interview plan, not a report of interviews already conducted. The team should replace this section with actual findings after collecting responses.

6.1 Interview Objectives

- Validate whether the proposed features address real needs of landlords and tenants.
- Identify the most frequent difficulties in managing and paying for rented rooms.
- Prioritize features for the first version of the system.

- Measure users' willingness to adopt a web application for rental management.

6.2 Target Participants

- **Landlords:** People who currently manage one or more rental properties.
- **Tenants:** People who currently rent a room or have recently completed a rental period.

The team should record participants' consent and avoid collecting unnecessary personal or financial information. Names and identifying details can be omitted from the report.

6.3 Recording and Analyzing Results

For each interview, record the participant category, date, key responses, recurring problems, requested features, and any concerns. Do not include identifiable information without permission.

After the interviews, group responses into themes such as billing, payments, contracts, maintenance, communication, reporting, usability, and privacy. Compare the findings from landlords and tenants, noting where their priorities agree or differ. Use the evidence to prioritize requirements rather than assuming that every proposed feature is equally important.